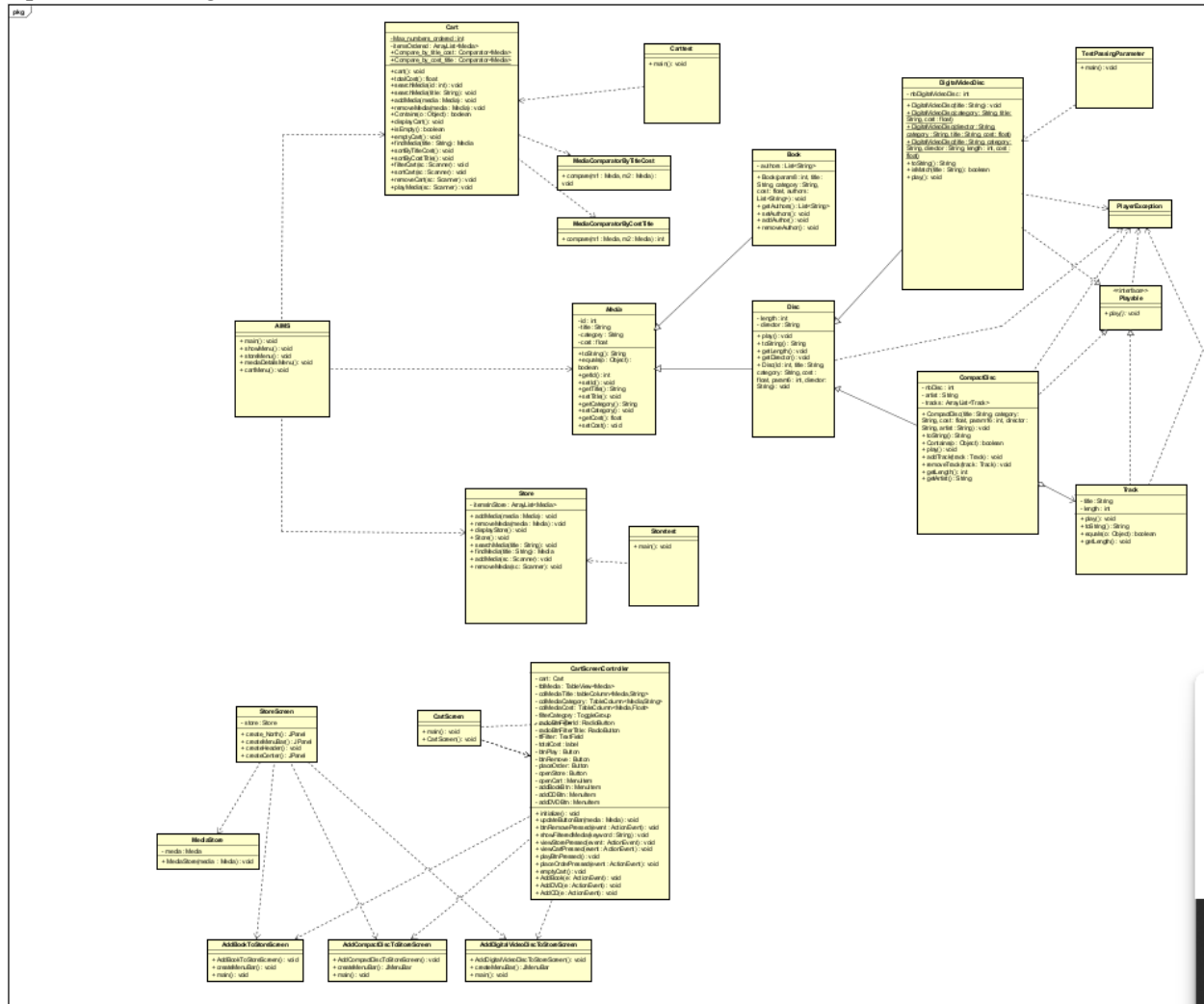
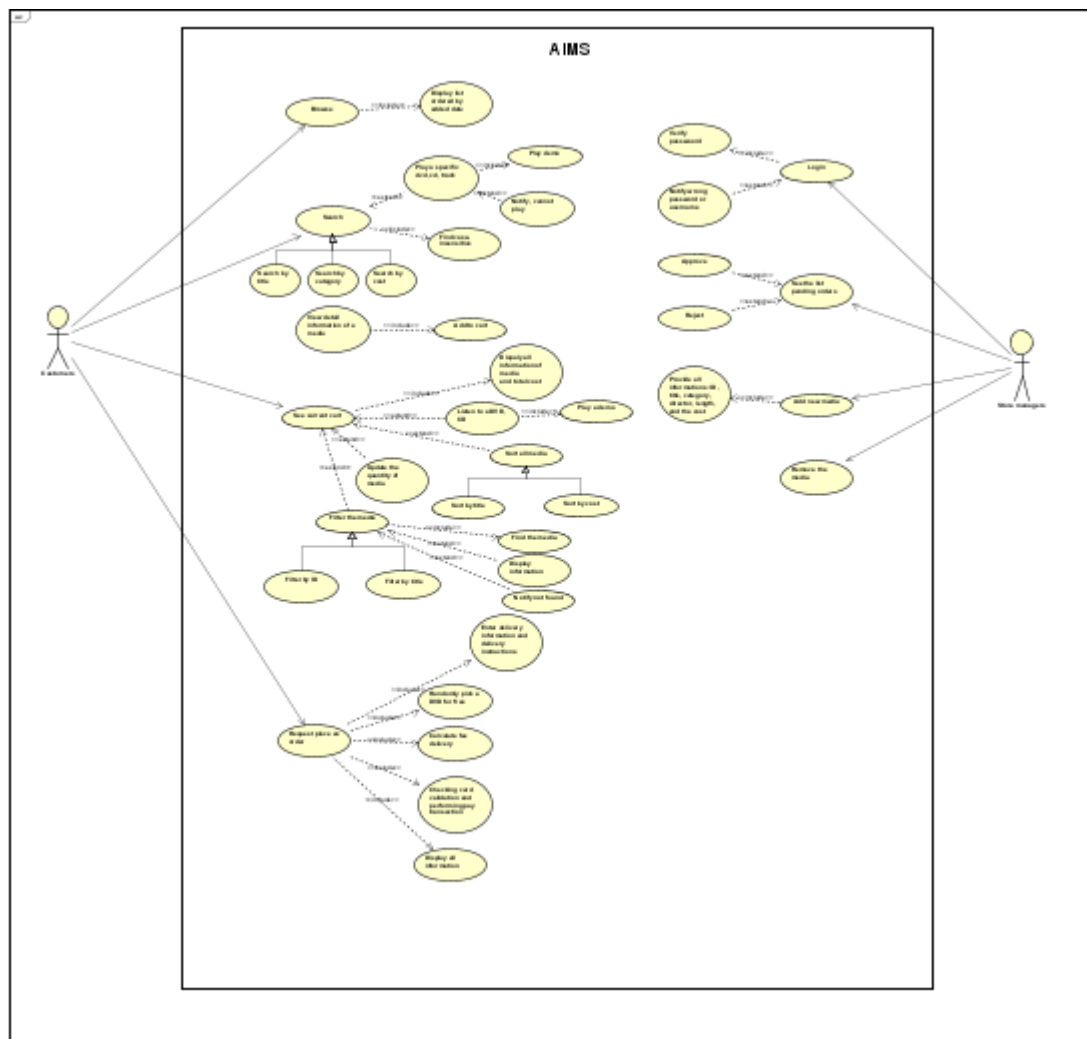


LAB 005: ANSWER

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Update Class Diagram





1. Top-Level Containers:

- **AWT:** AWT mainly uses the Frame class as its top-level container. The Frame corresponds to a standard window that comes with typical features like a title bar and buttons. However, AWT containers are considered heavyweight because they rely on the underlying operating system's windowing system for their appearance and behavior.
- **Swing:** Swing provides more flexible top-level containers such as JFrame, JWindow, JDialog, and JApplet. These are lightweight components, meaning they are entirely written in Java, offering greater customization and control. Swing also supports pluggable look-and-feel, allowing applications to adopt different visual styles and control how components are rendered.

2. Component Naming:

- **AWT:** In AWT, component names tend to be more generic, like Button, TextField, and Checkbox.
- **Swing:** Swing adopts a more descriptive naming convention for its components, such as JButton, JTextField, and JCheckBox. This makes the code clearer and easier to understand, improving readability and maintainability.

3. Event Handling:

- **AWT:** AWT uses a unified listener model where a single `EventListener` interface is used for handling all types of events. This can result in less organized and cluttered code as multiple events are managed through the same interface.
- **Swing:** Swing offers a more organized and specific event handling model. It uses distinct listener interfaces for different events, like `ActionListener` for button clicks and `MouseListener` for mouse actions. This separation helps keep the event-handling code cleaner and more modular.

4. Appearance and Look-and-Feel:

- **AWT:** The look and feel of AWT components are tied to the native graphical user interface (GUI) toolkit of the operating system. As a result, an AWT-based application may look different on various platforms.
- **Swing:** Swing offers a pluggable look-and-feel feature, enabling developers to customize the appearance of their applications. This allows Swing apps to appear consistent across different platforms or to mimic the look of the native OS GUI if desired.